Chapter 27: Problems with Solutions

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4. In the Bohr model of the hydrogen atom (which will MT be covered in detail in Chapter 42), an electron in the lowest energy state moves at a speed of 2.19×10^6 m/s in a circular path of radius 5.29×10^{-11} m. What is the effective current associated with this orbiting electron?

The period of the electron in its orbit is $T = 2\pi r/v$, and the current represented by the orbiting electron is

$$I = \frac{\Delta Q}{\Delta t} = \frac{|e|}{T} = \frac{v|e|}{2\pi r}$$

$$= \frac{(2.19\times10^{6}\ \text{m/s})(1.60\times10^{-19}\ \text{C})}{2\pi(5.29\times10^{-11}\ \text{m})}$$

$$= 1.05\times10^{-3}\ \text{C/s} = \boxed{1.05\ \text{mA}}$$

5. A proton beam in an accelerator carries a current of 125 μA. If the beam is incident on a target, how many protons strike the target in a period of 23.0 s?

If N is the number of protons, each with charge e, that hit the target in time Δt, the average current in the beam is $I = \Delta Q / \Delta t = Ne / \Delta t$, giving

$$N = \frac{I(\Delta t)}{e} = \frac{(125 \times 10^{-6}\ \text{C/s})(23.0\ \text{s})}{1.60 \times 10^{-19}\ \text{C/proton}} = \boxed{1.80 \times 10^{16}\ \text{protons}}$$

2. A small sphere that carries a charge q is whirled in a circle at the end of an insulating string. The angular frequency of revolution is ω. What average current does this revolving charge represent?

The period of revolution for the sphere is $T = \frac{2\pi}{\omega}$, and the average current represented by this revolving charge is $I = \frac{q}{T} = \boxed{\frac{q\omega}{2\pi}}$.

7. Suppose the current in a conductor decreases exponentially with time according to the equation $I(t) = I_0 e^{-t/\tau}$, where I_0 is the initial current (at $t = 0$) and τ is a constant having dimensions of time. Consider a fixed observation point within the conductor. (a) How much charge passes this point between $t = 0$ and $t = \tau$? (b) How much charge passes this point between $t = 0$ and $t = 10\tau$? (c) **What If?** How much charge passes this point between $t = 0$ and $t = \infty$?

From $I = \frac{dQ}{dt}$, we have $dQ = I\,dt$.

From this, we derive the general integral: $Q = \int dQ = \int I\,dt$

In all three cases, define an end-time, T: $Q = \int_0^T I_0 e^{-t/\tau} dt$

Integrating from time $t = 0$ to time $t = T$: $Q = \int_0^T \left(-I_0\tau\right) e^{-t/\tau} \left(-\frac{dt}{\tau}\right)$

We perform the integral and set $Q = 0$ at $t = 0$ to obtain

$$Q = -I_0\tau\left(e^{-T/\tau} - e^0\right) = I_0\tau\left(1 - e^{-T/\tau}\right)$$

(a) If $T = \tau$: $Q(\tau) = I_0\tau\left(1 - e^{-1}\right) = \boxed{(0.632)\,I_0\tau}$

(b) If $T = 10\tau$: $Q(10\tau) = I_0\tau\left(1 - e^{-10}\right) = \boxed{(0.999\,95)\,I_0\tau}$

(c) If $T = \infty$: $Q(\infty) = I_0\tau\left(1 - e^{-\infty}\right) = \boxed{I_0\tau}$

9. The quantity of charge q (in coulombs) that has passed through a surface of area 2.00 cm^2 varies with time according to the equation $q = 4t^3 + 5t + 6$, where t is in seconds. (a) What is the instantaneous current through the surface at $t = 1.00$ s? (b) What is the value of the current density?

We are given $q = 4t^3 + 5t + 6$. The area is

$$A = (2.00\ \text{cm}^2)\left(\frac{1.00\ \text{m}}{100\ \text{cm}}\right)^2 = 2.00 \times 10^{-4}\ \text{m}^2$$

(a) $$I(1.00\ \text{s}) = \left.\frac{dq}{dt}\right|_{t=1.00\ \text{s}} = \left.(12t^2 + 5)\right|_{t=1.00\ \text{s}} = \boxed{17.0\ \text{A}}$$

(b) $$J = \frac{I}{A} = \frac{17.0\ \text{A}}{2.00 \times 10^{-4}\ \text{m}^2} = \boxed{85.0\ \text{kA/m}^2}$$

12. An electric current in a conductor varies with time according to the expression $I(t) = 100 \sin(120\pi t)$, where I is in amperes and t is in seconds. What is the total charge passing a given point in the conductor from $t = 0$ to $t = \frac{1}{240}$ s?

To find the total charge passing a point in a given amount of time, we use $I = \frac{dq}{dt}$, from which we can write

$$q = \int dq = \int I\,dt = \int_0^{1/240\ s} (100\ A)\sin\left(\frac{120\pi t}{s}\right)dt$$

$$q = \frac{-100\ C}{120\pi}\left[\cos\left(\frac{\pi}{2}\right) - \cos 0\right] = \frac{+100\ C}{120\pi} = \boxed{0.265\ C}$$

14. A lightbulb has a resistance of 240 Ω when operating **W** with a potential difference of 120 V across it. What is the current in the lightbulb?

From Equation 27.7, we obtain

$$I = \frac{\Delta V}{R} = \frac{120\ \text{V}}{240\ \Omega} = 0.500\ \text{A} = \boxed{500\ \text{mA}}$$

16. A 0.900-V potential difference is maintained across a 1.50-m length of tungsten wire that has a cross-sectional area of 0.600 mm^2. What is the current in the wire?

$\Delta V = IR$ and $R = \frac{\rho \ell}{A}$. The area is

$$A = (0.600\ \text{mm}^2)\left(\frac{1.00\ \text{m}}{1\,000\ \text{mm}}\right)^2 = 6.00 \times 10^{-7}\ \text{m}^2$$

From the potential difference, we can solve for the current, which gives

$$\Delta V = \frac{I\rho\ell}{A} \rightarrow I = \frac{\Delta V A}{\rho \ell} = \frac{(0.900\ \text{V})(6.00 \times 10^{-7}\ \text{m}^2)}{(5.60 \times 10^{-8}\ \Omega \cdot \text{m})(1.50\ \text{m})}$$

$$I = \boxed{6.43\ \text{A}}$$

21. A portion of Nichrome wire of radius 2.50 mm is to be used in winding a heating coil. If the coil must draw a current of 9.25 A when a voltage of 120 V is applied across its ends, find (a) the required resistance of the coil and (b) the length of wire you must use to wind the coil.

(a) From the definition of resistance,

$$R = \frac{\Delta V}{I} = \frac{120\ \text{V}}{9.25\ \text{A}} = \boxed{13.0\ \Omega}$$

(b) The resistivity of Nichrome (from Table 27.2) is $1.50 \times 10^{-6}\ \Omega \cdot \text{m}$.

We find the length of wire from

$$R = \frac{\rho \ell}{A} = \frac{\rho \ell}{\pi r^2}$$

solving for the length ℓ gives

$$\ell = \frac{R \pi r^2}{\rho} = \frac{(13.0\ \Omega)\pi\left(2.50 \times 10^{-3}\ \text{m}\right)^2}{\left(1.50 \times 10^{-6}\ \Omega \cdot \text{m}\right)} = \boxed{170\ \text{m}}$$

26. A certain lightbulb has a tungsten filament with a resistance of 19.0 Ω when at 20.0°C and 140 Ω when hot. Assume the resistivity of tungsten varies linearly with temperature even over the large temperature range involved here. Find the temperature of the hot filament.

$R = R_0[1 + \alpha(\Delta T)]$ gives

$$140\ \Omega = (19.0\ \Omega)\left[1 + \left(4.50 \times 10^{-3}/°C\right)\Delta T\right]$$

Solving,

$$\Delta T = 1.42 \times 10^{3}\ °C = T - 20.0°C$$

And the final temperature is $\boxed{T = 1.44 \times 10^{3}\ °C}$

27. What is the fractional change in the resistance of an iron filament when its temperature changes from 25.0°C to 50.0°C?

If we ignore thermal expansion, the change in the material's resistivity with temperature $\rho = \rho_0[1 + \alpha\Delta T]$ implies that the change in resistance is $R - R_0 = R_0\alpha\Delta T$. The fractional change in resistance is defined by $f = (R - R_0)/R_0$. Therefore,

$$f = \frac{R_0\alpha\Delta T}{R_0} = \alpha\Delta T = \left(5.00\times10^{-3}\ ^\circ\text{C}^{-1}\right)(50.0^\circ\text{C} - 25.0^\circ\text{C}) = \boxed{0.12}$$

29. If a certain silver wire has a resistance of 6.00 Ω at 20.0°C, what resistance will it have at 34.0°C?

$$R = R_0\left[1 + \alpha\left(T - T_0\right)\right]$$

$$= (6.00\ \Omega)\left[1 + \left(3.8\times10^{-3}\ (^\circ\text{C})^{-1}\right)(34.0^\circ\text{C} - 20.0^\circ\text{C})\right]$$

$$= \boxed{6.32\ \Omega}$$

31. M (a) A 34.5-m length of copper wire at 20.0°C has a radius of 0.25 mm. If a potential difference of 9.00 V is applied across the length of the wire, determine the current in the wire. (b) If the wire is heated to 30.0°C while the 9.00-V potential difference is maintained, what is the resulting current in the wire?

(a) The resistance at 20.0°C is

$$R_0 = \frac{\rho \ell}{A} = \frac{(1.7\times10^{-8}\ \Omega\cdot\text{m})(34.5\ \text{m})}{\pi(0.25\times10^{-3}\ \text{m})^2} = 2.99\ \Omega$$

and the current is

$$I = \frac{\Delta V}{R_0} = \frac{9.00\ \text{V}}{3.00\ \Omega} = \boxed{3.01\ \text{A}}$$

(b) At 30.0°C, from Equation 27.20,

$$R = R_0[1+\alpha(\Delta T)]$$

$$= (2.99\ \Omega)\left[1+\left(3.9\times10^{-3}\ (°\text{C})^{-1}\right)(30.0°\text{C}-20.0°\text{C})\right] = 3.10\ \Omega$$

The current is then

$$I = \frac{\Delta V}{R_0} = \frac{9.00\ \text{V}}{3.10\ \Omega} = \boxed{2.90\ \text{A}}$$

37. In a hydroelectric installation, a turbine delivers 1 500 hp to a generator, which in turn transfers 80.0% of the mechanical energy out by electrical transmission. Under these conditions, what current does the generator deliver at a terminal potential difference of 2 000 V?

$$P = 0.800(1\,500\ \text{hp})(746\ \text{W/hp}) = 8.95 \times 10^5\ \text{W}$$

Then, from $P = I\Delta V$,

$$I = \frac{P}{\Delta V} = \frac{8.95 \times 10^5\ \text{W}}{2\,000\ \text{V}} = \boxed{448\ \text{A}}$$

39. A certain waffle iron is rated at 1.00 kW when connected to a 120-V source. (a) What current does the waffle iron carry? (b) What is its resistance?

(a) From Equation 27.21,

$$P = I\Delta V \rightarrow I = P/\Delta V = \left(1.00 \times 10^3 \text{ W}\right)/(120 \text{ V}) = \boxed{8.33 \text{ A}}$$

(b) From Equation 27.23,

$$P = \Delta V^2/R \rightarrow R = \Delta V^2/P = (120 \text{ V})^2/\left(1.00 \times 10^3 \text{ W}\right) = \boxed{14.4\ \Omega}$$

45. Batteries are rated in terms of ampere-hours (A · h). **W** For example, a battery that can produce a current of 2.00 A for 3.00 h is rated at 6.00 A · h. (a) What is the total energy, in kilowatt-hours, stored in a 12.0-V battery rated at 55.0 A · h? (b) At \$0.110 per kilowatt-hour, what is the value of the electricity produced by this battery?

(a) The total energy stored in the battery is

$$\Delta U_E = q(\Delta V) = It(\Delta V)$$

$$= (55.0\ \text{A}\cdot\text{h})(12.0\ \text{V})\left(\frac{1\ \text{C}}{1\ \text{A}\cdot\text{s}}\right)\left(\frac{1\ \text{J}}{1\ \text{V}\cdot\text{C}}\right)\left(\frac{1\ \text{W}\cdot\text{s}}{1\ \text{J}}\right)$$

$$= 660\ \text{W}\cdot\text{h} = \boxed{0.660\ \text{kWh}}$$

(b) The value of the electricity is

$$\text{Cost} = (0.660\ \text{kWh})\left(\frac{\$0.110}{1\ \text{kWh}}\right) = \boxed{\$0.072\ 6}$$

46. Residential building codes typically require the use
W of 12-gauge copper wire (diameter 0.205 cm) for wiring receptacles. Such circuits carry currents as large as 20.0 A. If a wire of smaller diameter (with a higher gauge number) carried that much current, the wire could rise to a high temperature and cause a fire. (a) Calculate the rate at which internal energy is produced in 1.00 m of 12-gauge copper wire carrying 20.0 A. (b) **What If?** Repeat the calculation for a 12-gauge aluminum wire. (c) Explain whether a 12-gauge aluminum wire would be as safe as a copper wire.

(a) The resistance of 1.00 m of 12-gauge copper wire is

$$R = \frac{\rho \ell}{A} = \frac{\rho \ell}{\pi (d/2)^2} = \frac{4\rho \ell}{\pi d^2} = \frac{4\left(1.7 \times 10^{-8}\ \Omega \cdot \text{m}\right)(1.00\ \text{m})}{\pi \left(0.205 \times 10^{-2}\ \text{m}\right)^2}$$

$$= 5.2 \times 10^{-3}\ \Omega$$

The rate of internal energy production is

$$P = I\Delta V = I^2 R = (20.0\ \text{A})^2 \left(5.2 \times 10^{-3}\ \Omega\right) = \boxed{2.1\ \text{W}}$$

(b) $$R = \frac{4\rho \ell}{\pi d^2} = \frac{4\left(2.82 \times 10^{-8}\ \Omega \cdot \text{m}\right)(1.00\ \text{m})}{\pi \left(0.205 \times 10^{-2}\ \text{m}\right)^2} = 8.54 \times 10^{-3}\ \Omega$$

$$P = I\Delta V = I^2 R = (20.0\ \text{A})^2 \left(8.54 \times 10^{-3}\ \Omega\right) = \boxed{3.42\ \text{W}}$$

(c) It would not be as safe. If surrounded by thermal insulation, it would get much hotter than a copper wire.

49. A coil of Nichrome wire is 25.0 m long. The wire has a diameter of 0.400 mm and is at 20.0°C. If it carries a current of 0.500 A, what are (a) the magnitude of the electric field in the wire and (b) the power delivered to it? (c) **What If?** If the temperature is increased to 340°C and the potential difference across the wire remains constant, what is the power delivered?

First, we compute the resistance of the wire:

$$R = \frac{\rho \ell}{A} = \frac{\left(1.50 \times 10^{-6}\ \Omega \cdot \text{m}\right) 25.0\ \text{m}}{\pi \left(0.200 \times 10^{-3}\ \text{m}\right)^2} = 298\ \Omega$$

The potential drop across the wire is then

$$\Delta V = IR = (0.500\ \text{A})(298\ \Omega) = 149\ \text{V}$$

(a) The magnitude of the electric field in the wire is

$$E = \frac{\Delta V}{\ell} = \frac{149\ \text{V}}{25.0\ \text{m}} = \boxed{5.97\ \text{V/m}}$$

(b) The power delivered to the wire is

$$P = (\Delta V) I = (149\ \text{V})(0.500\ \text{A}) = \boxed{74.6\ \text{W}}$$

(c) We use Equation 27.20 and Table 27.2:

$$R = R_0\left[1 + \alpha\left(T - T_0\right)\right] = (298\ \Omega)\left[1 + \left(0.400 \times 10^{-3}/^{\circ}\text{C}\right) 320^{\circ}\text{C}\right]$$
$$= 337\ \Omega$$

To find the power delivered, we first compute the current flowing through the wire:

$$I = \frac{\Delta V}{R} = \frac{149\ \text{V}}{337\ \Omega} = 0.443\ \text{A}$$

then,

$$P = (\Delta V) I = (149\ \text{V})(0.443\ \text{A}) = \boxed{66.1\ \text{W}}$$